

SchuF Globe Control Valves



SchuF 

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*SchuF is fully registered,
accredited and certified
worldwide*

SchuF Worldwide Group

SchuF is an industry-renowned valve supplier with over 100 years experience designing and manufacturing application-specific valve solutions.

SchuF has developed over 20,000 control valve variations in its hundred-year history. Each has its own specific characteristics tailored to the process control elements that are most important for it – pressure, level, flow or temperature.

SchuF has the capability to ship our unique and highly-praised valve solutions worldwide from design and production facilities located in Germany, India, Ireland, Italy, the United Kingdom

and the United States, where SchuF's skilled team of engineers and product specialists design each valve from the ground up to meet specific application requirements and provide optimal service life and performance.

As well as a complete range of globe control valve designs, SchuF also supply a wide range of other bespoke valve types, as well as both pneumatic diaphragm and pneumatic piston spring-return actuators.



SchuF Globe Control Valve Products- Introduction

Individual process issues: bespoke valve solutions

SchuF Globe Valves are ideal for severe operating conditions with high flow rates, high pressure or temperature drops, corrosive & erosive substances, vibrations, or where crystallisation can be an issue. SchuF has developed specialist solutions to many associated design challenges across numerous globe control valve applications.

Each design has its own specific characteristics tailored to the process control elements that are most critical.



Gas Field Development

- On-shore facility, Middle-East
- 8" ASME Class 300



Sour Gas Service on Wellhead Facility

- Electro-hydraulic Actuator
- 4" ASME Class 2500



Gas Service

- Extended Bonnet
- Diaphragm / Spring-Return
- or
- ASME Class 300

Cryogenic Service

- Extended Bonnet
- Diaphragm / Spring-Return Actuator



Boiler Feed-water Paper Factory

- 3" ASME Class 900
- 1.5" ASME Class 600
- Diaphragm Spring-return Actuator



High-Pressure Hydro-cracking

- 4" ASME Class 2500
- Multistage
- Double-acting Pneumatic Piston Actuator

Control Valve Product Range

Description

The SchuF range of Control Valves has been designed to meet the ever increasing demands of modern processing requirements in the Oil & Gas, Power, Petrochemical, Chemical and general industries. SchuF valves include a wide range of trim designs, from simple contoured plugs or single-stage drilled-hole cages through to the labyrinth disc-stack designs, which will meet the requirements of critical and / or demanding applications (see pages 18, 19, 20). Valves are available as globe valves (**Series-G**) or angle valves (**Series-A**), incorporating identical trim design and options.

Body Size

- 25 mm to 600 mm
- 1 in to 24 in

Pressure Class

- ASME CL150 to ASME CL4500
- BS EN 1092-1 PN16 to PN 100

End Connections

- Flanged
- RTJ
- Butt weld end
- Hubbed (Clamped)

Design Standards

- ASME B16.34
- ASME B16.5 (Pipe Flanges)
- ASME B16.25 (Butt Weld End)
- API 6A
- ASME VIII
- NACE MR-01-75
- PED 97/23/EC

Bonnet Designs

- Standard
- Extended
- Bellow-Sealed
- Cryogenic

Trim Design *(see images, right)*

Contour Trim

This trim would be used for low pressure drop applications (see Figure 1). This is a plug with a profiled face to give the required flow characteristic.

Spline Trim

Figure 2 presents the trim which would be used for low flow conditions. This trim has a Cv range from 5 down to 0.01. Both single and multistage designs are available.

VC-1 Trim

This is a Cage Guided Trim (Figure 3) and has low pressure recovery which means it can handle much higher pressure-drops than the Contour Trim before cavitation occurs. The standard trim design ranges from VC-1, a single-stage cage design, through to VC-5, a 5-stage design, for higher pressure drops, controlling velocity and noise. Figure 4 presents a VC-3 arrangement.

Stacker Trim

For the more severe service applications, the Stacker Trim (Fig. 6), which is a labyrinth-type trim design with up to 36 turns (stages of pressure drop), may be required- see page 20.

Body Materials

SchuF control valves can be manufactured in most forms of castable alloy, such as;

- Carbon Steel (WCB, WCC, LCC)
- Stainless Steel (CF8M, CF3M, etc.)
- Chrome Moly (WC6, WC9, C12A)
- Duplex St.St. (A995 4A / 5A /6A)
- High Nickel - Monel®, Alloy 625
- Hastelloy®
- Inconel Alloy 740H (UNS N07740)
- Aluminium Bronze
- Titanium

Trim Materials

Trim material selected will be compatible with the specified body material, see Figures 13 to 15 for standard combinations. In addition to the base trim materials, Gr. 6 Stellite / Ultimet overlays and solid tungsten carbide inserts may also be specified in order to prevent erosion damage. on contaminated services.

Control Valve Trim Designs

Fig. 1 - Contoured Trim, with Ported Cage

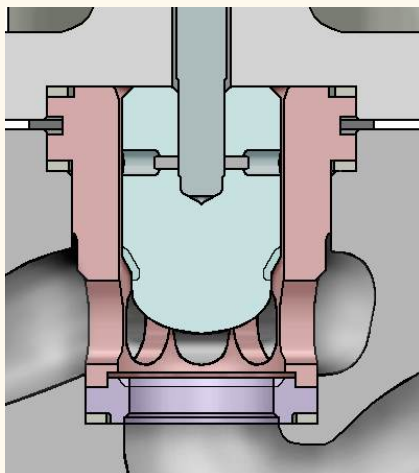


Fig. 2 - Needle Spline Trim

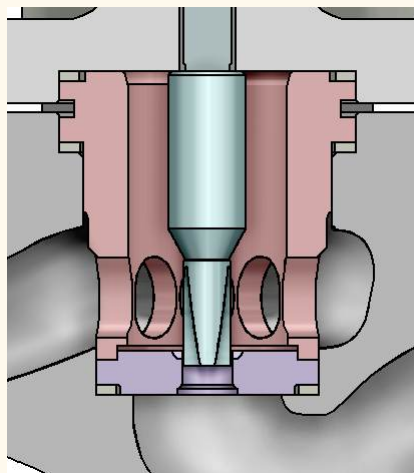


Fig. 3 - Cage-Guided Trim (VC-1)

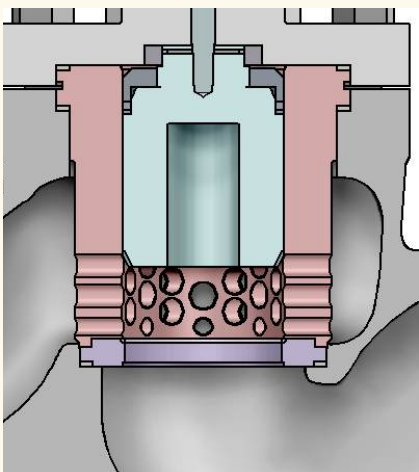


Fig. 4 - Three-Stage Trim (VC-3)

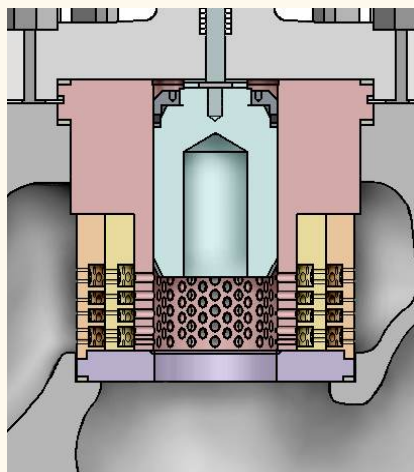


Fig. 5 - Seat-Guided Trim (VC-1)

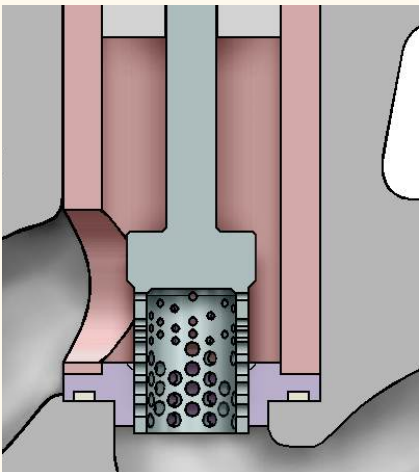
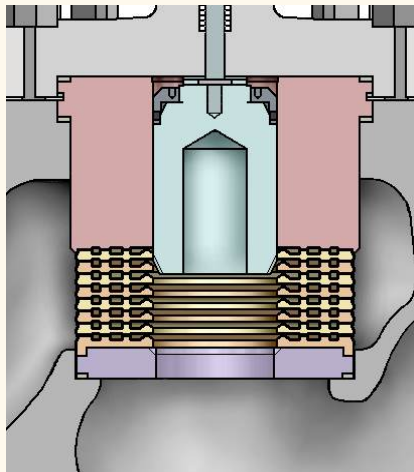


Fig. 6 - Stacker Trim



Trim Features and Benefits

Fig. 5 - Single-Stage Cage-Guided Trim (VC-1)

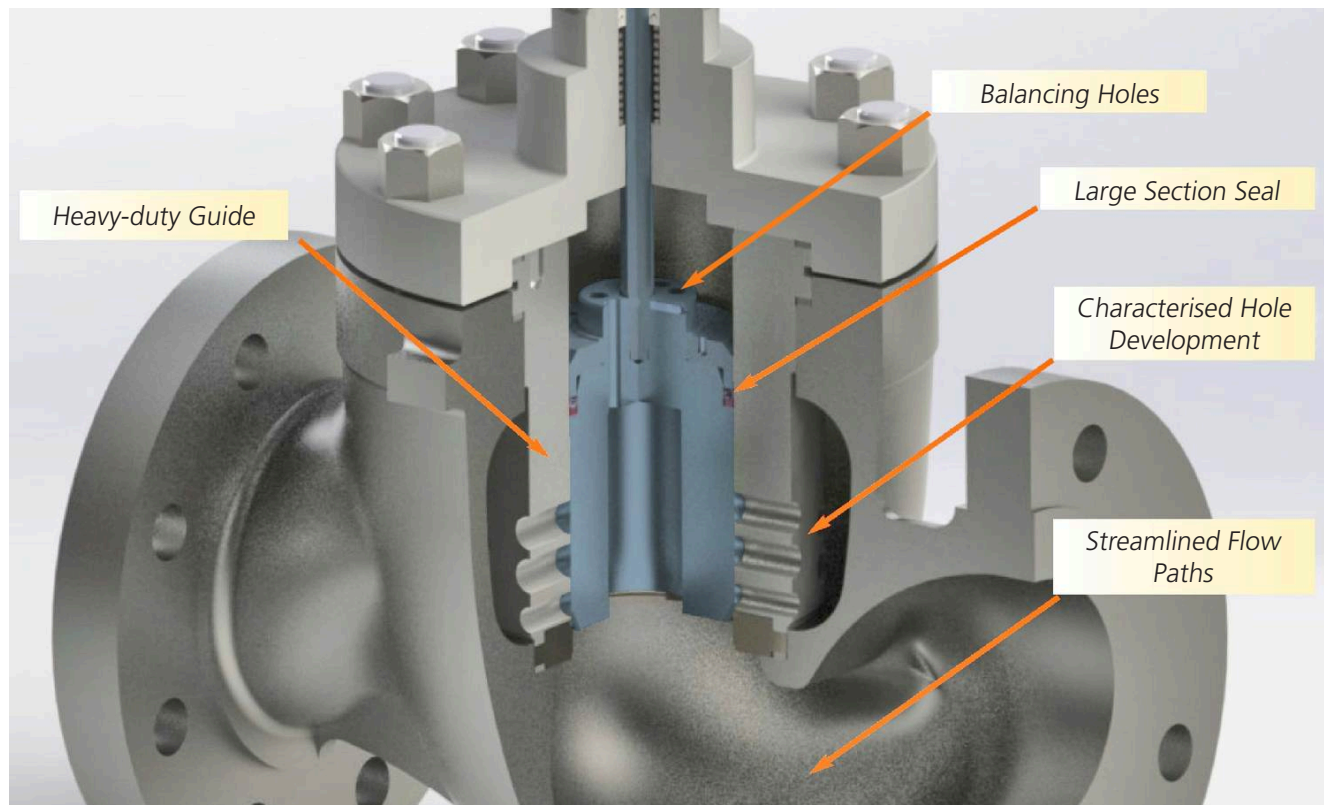
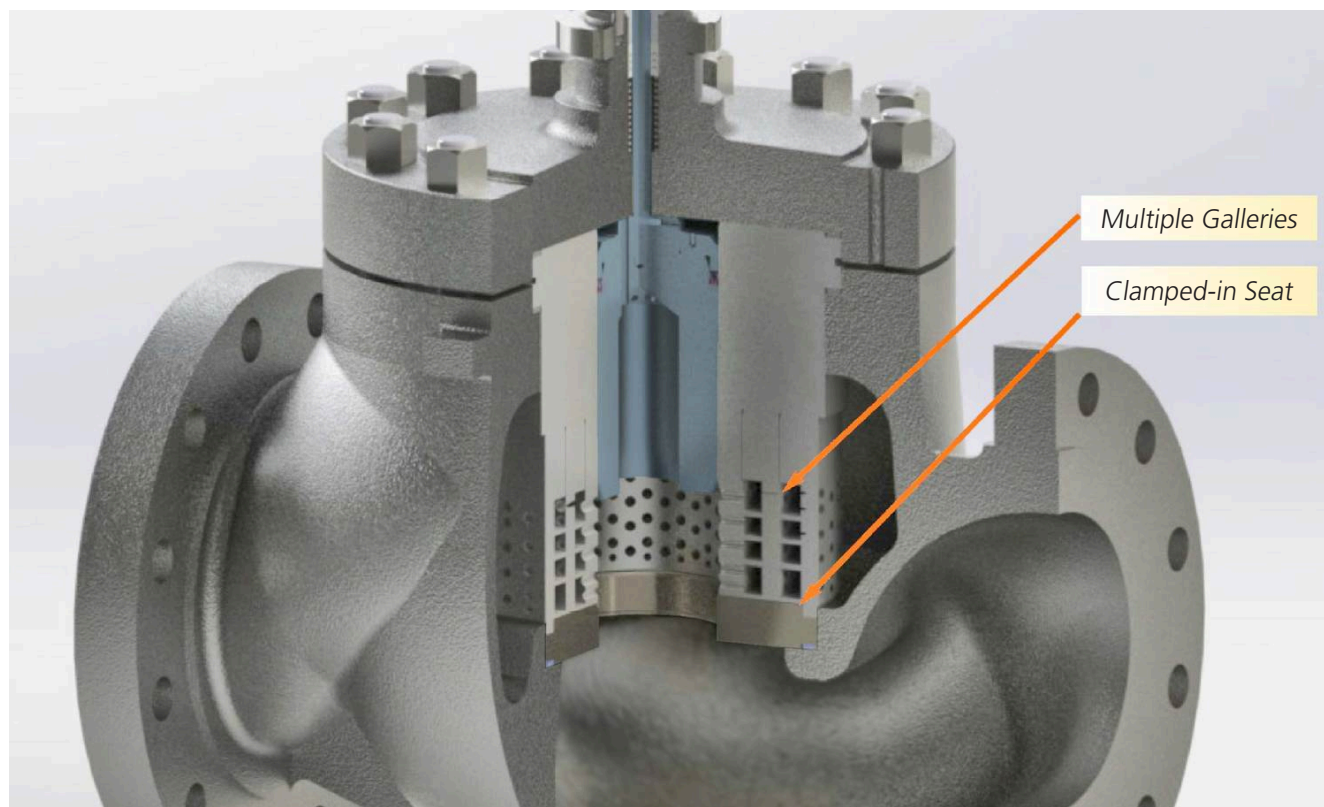


Figure 6 - Three-Stage Guided Trim - (VC-3)



Control Valve Trim- Features

Features and their Benefits

Multiple galleries.	→	Fully-apportioned pressure-drop through full travel.
Large number of flow paths through cage.	→	Lower noise levels.
Minimal plug / guide clearance.	→	High rangeability / improved plug stability.
Characterised hole development.	→	Stable control / higher rangeability
Large deadband on erosive applications.	→	Extended trim life.
Balanced trim, large section seal or damping strip	→	Reduced unbalanced force, improved guiding / minimise vibration issues.
Configured variable stage trim for specific application.	→	Reduced valve size, reduced dynamic forces.
Clamped-in guide and seat.	→	Ease of maintenance.
Well proven body/bonnet bolting system.	→	High integrity sealing.
Streamlined flow passages.	→	Reduced valve size.
Supplementary noise control components	→	Better control of noise issues.

Design Integrity

- Clamped cage for positive guiding on severe service applications.
- High integrity plug guiding system.
- Low emission packings i.e. API 522.
- ASME VIII Body/Bonnet bolting.

Quality-Assured Manufacturing

- Rigorously tested to ensure specified performance on site.
- Quality assurance systems in accordance with ISO 9001:2008. LRQ4005817
- Optional full ISO 15156 / NACE MR-01-75 certification.
- PED Certified.

Right: DN200
Globe Control Valve
with integral Heating Jacket



Typical Applications

Oil & Gas: MOL pump pressure control; MOL Pump minimum flow re-cycle; Overboard dump valve; High-pressure seawater control systems; Minimum-flow pump recycle; Separator level control; Compressor recycle / Anti-surge valves; Gas blow-down

Power: Boiler feedwater pressure-control valve, Condensate minimum-flow recycle, Emergency drain valve, High-pressure steam blowdown, Spray-water control.

Control Valve Case Study- From Process to Proposal

A case study showing how SchuF analyse a customer's requirements to engineer best-case solutions

Customer Process

The process is a tube reactor system, where the reaction takes place in the pipeline with additives injected at various points. Good controllability at low flow during startup is required, and any increase in flow has to be steady without fluctuations. An existing valve with Ti Gr.5 trim required annual replacement of the plug, seat and stem due to erosion. In addition, particles in the flow can cause blockages.

■ **Flashing Service: normal recommendations**

It is normally good practice on flashing duties to use a 'flow-over' design, and also, where possible, to utilise an angle valve to avoid body erosion and

potential erosion of the pressure boundary wall. In cases where a globe valve design is used, it would normally be recommended that some degree of body protection be utilised; hardened trim components will minimise erosion.

■ **Flashing and Multi-Stage Trims**

Multi-stage trims on applications with a high degree of flash are not necessarily the best solution. If it is possible to get the flashing to occur across the final stage, then a multi-stage trim can be acceptable. However, if the flashing occurs as soon as the pressure begins to fall, this can lead to an even more erosive fluid impacting the subsequent stages, leading to accelerated erosion damage.

Basis of Proposed Design

It would seem sensible to try and maintain a similar design (i.e. single-stage) as previously used by the customer- however, changing the valve design from an angle to a globe could introduce a potential for erosion of the pressure boundary. The starting pressure drop is greater than SchuF would normally recommend for single-stage trim: However, this situation is of short duration only.

■ **Flow Direction**

The shape of the existing globe casting makes it marginally easier to protect the body with a 'flow-under' design together with a protective sleeve rather than a 'flow-over' design with a protective seat diffuser.

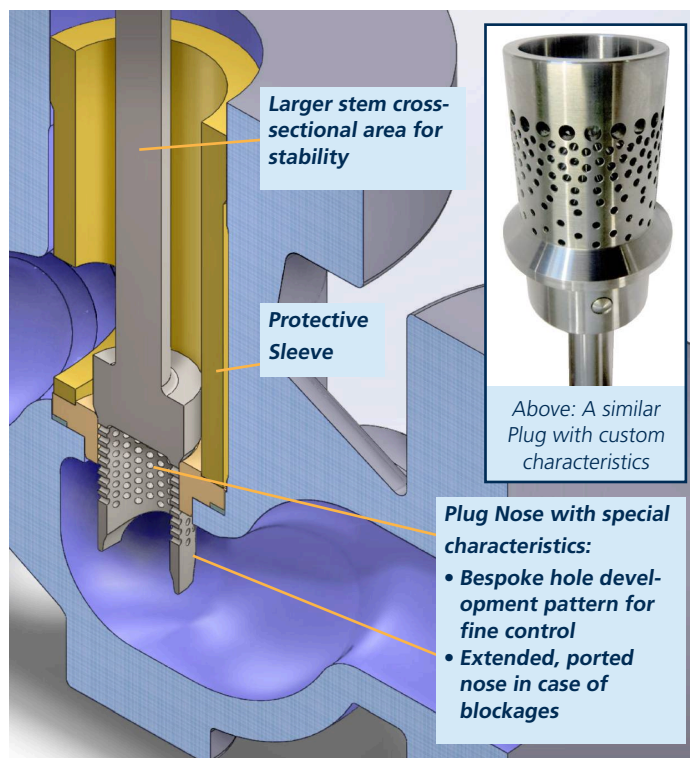
■ **Trim Design**

In order to distribute the flow into the outer protective sleeve, SchuF recommended an extended plug nose design, with radially drilled holes, which operate during up to 40 mm of valve travel. Above this there would be 4 off ports, available with extended actuator travel (to, say, 60 mm) in case of blockage of the radially drilled holes. The plug nose hole development would be designed to give a modified Eq% flow characteristic, giving controllability at low valve openings (see image above).

Proposed Trim

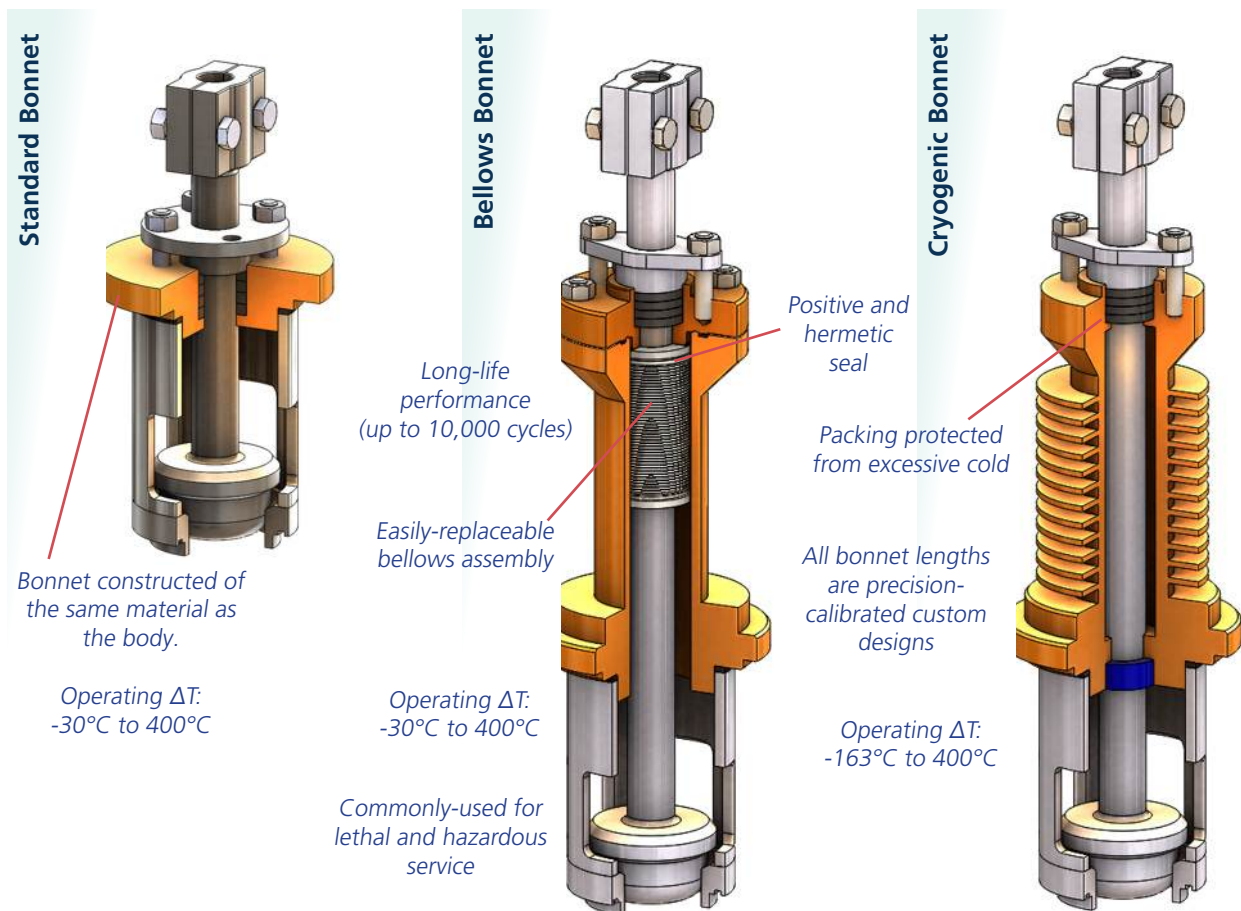
■ **Protective Sleeve**

The protective sleeve has been designed to make use of the available space within the valve gallery.

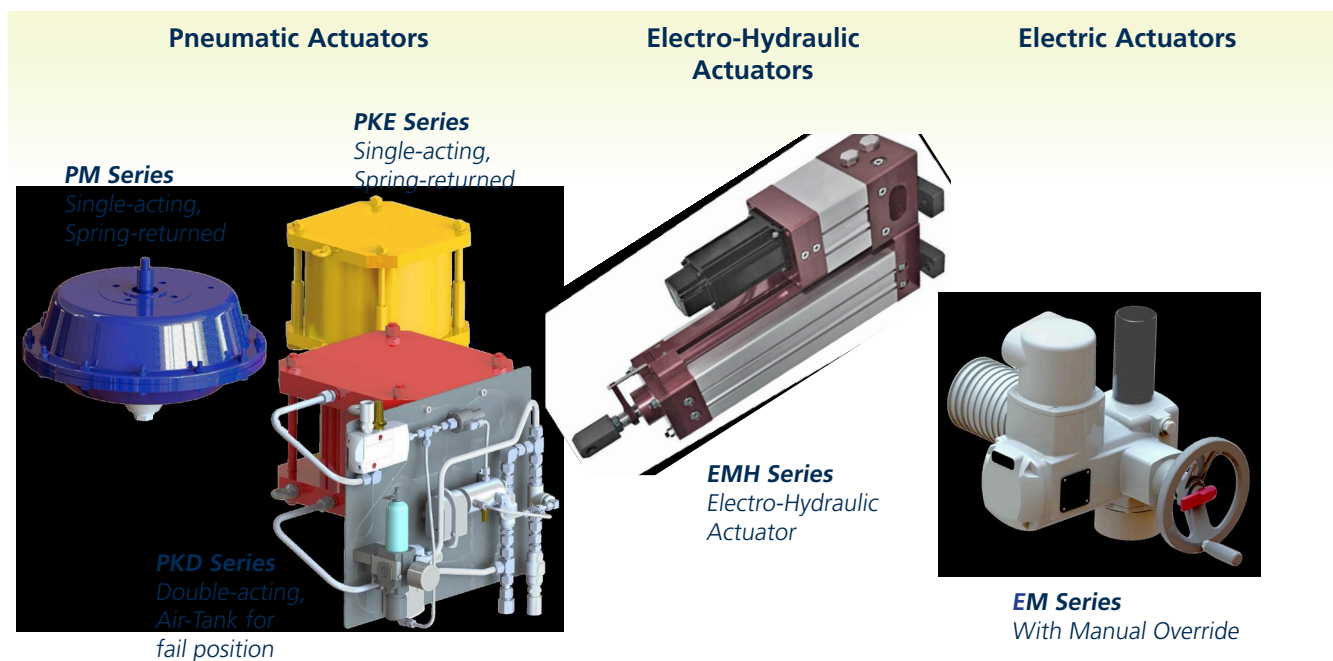


There is a radial space between the plug head outer diameter and the inner diameter of the protective sleeve. This is as large as is practical, bearing in mind the Cv required, and the need to protect the body wall. Within the protective sleeve there is space for the flow to expand; as the fluid flashes, the mixture's specific volume will increase, leading to higher velocities. The flow is then directed from the protective sleeve through a large port, sized as close as possible to the valve body dimensions.

Control Globe Valve Bonnet Selection



Control Globe Valve Actuation



Technical Data

The following tables present the limits for various flow parameters such as pressure drops, flow velocities and operating temperatures, as well as component material selection.

Figure 7 - Maximum Stage Pressure Drops

Trim Design	Flow Direction	Liquid Flows				Gas Flows			
		Clean		Contaminated		Clean		Contaminated / Wet	
		bar	psi	bar	psi	bar	psi	bar	psi
VC-1	Over	40	580	25	360	100	1450	25	360
	Under	10	145	-	-	100	1450	-	-
VC-2	Over	60	870	40	580	110	1595	50	725
	Under	60	870	-	-	150	2170	-	-
VC-3	Over	90	1300	60	870	120	1740	80	1160
	Under	90	1300	100	1450	180	2610	100	1450
VC-4	Over	120	1740	80	1160	160	2320	100	1450
	Under	140	2030	100	1450	210	3050	150	2170
VC-5	Over	160	2320	100	1450	200	2900	120	1740
	Under	190	2750	100	1459	230	3330	180	2610

Notes

1. On flashing applications other considerations may affect trim and material overlay options.
2. These limits apply on the basis that cavitation has been eliminated.
3. On liquid applications, where final stage pressure drops are greater than 50 bar, angle body valves should be recommended.
4. Where entry is blank, valve would never be used under stated conditions.

Figure 8 - Maximum Allowable Stage pressure Drop Limits for Given Materials

Trim Material / Overlay / Insert	Typical Hardness Rockwell 'C'	Max. Δp per stage - Liquid	
		Bar	PSI
316L St. St.	183 (BHN)	20	290
17.4 PH	29 - 33	30	435
Monel	205 (BHN)	30	435
Duplex St. St.	205 (BHN)	30	435
Stellite® Gr. 6 Face	39 - 43	35	500
Full Stellite® Gr. 6	39 - 43	40	580
Full Stellite® Gr. 21	27 - 40	40	580
Hardened 440 St. St.	45	40	580
Tungsten Carbide	80	> 60	> 870

Figure 9 - Recommended Maximum Velocities for Liquid Service

Valve Type	Valve Rating	Trim Style	Valve Size		Maximum Velocity					
					Carbon Steel		Alloy Steel		Bronze, Cu/Ni	
			mm	inches	m/s	ft/s	m/s	ft/s	m/s	ft/s
Globe/ Angle	All	Contour	15 - 20 80 - 150	0.5 - 2 3 - 6	12.5 10.4	41 34	14 10.4	46 34	7.6 6.2	25 20
	All	Ports	25 - 300 350 - 500	1 - 12 14 - 20	13.1 10.7	43 35	16 13	52 43	7.9 6.4	26 21
Globe	All	VC-1	25 - 300 350 - 500	1 - 12 14 - 20	13 11	43 35	16 13	52 43	8 6.5	26 21
Angle	All	VC-1	25 - 300 350 - 500	1 - 12 14 - 20	15 12.2	48 40	17 14	57 48	8.8 7	29 24

Technical Data

Figure 10 - Recommended Maximum Velocities for Gas/Vapour

Valve Type	Valve Rating	Trim Style	Valve Size		Max. Inlet Velocity		Max. Outlet Velocity		Max. Outlet Mach No. For Required Noise Level		
			mm	inches	m/s	ft/s	m/s	ft/s	> 95 dBA	< 95 dBA	< 85 dBA
Globe/ Angle	Up to and including ASME 600	Contour	15 - 50	0.5 - 2	104	340	253	830	0.65	0.5	0.3
	All	Ports	25 - 300 350 - 500	1 - 12 14 - 20	144	475					
Cage	All	VC-1	25 - 300 350 - 500	1 - 12 14 - 20	144	475	253	830	0.65	0.5	0.3
Angle	All	VC-1	25 - 300 350 - 500	1 - 12 14 - 20	144	475	Sonic		0.65	0.5	0.3

The following tables give the selection criteria based on fluid temperature and the actual duty for which the valve is to be used. For example, bonnet and packing selection, plug seal and the trim materials of construction.

Figure 11 - Bonnet / Packing Selection For Applicable Temperature Ranges

Component ↓ Applicable Temperature Range →	< -100°C	-100°C to -46°C	-46°C to 250°C	250°C to 400°C	≥ 400°C
	< -150°F	-150°F to -46°F	50°F to 480°F	450°F to 750°F	≥ 752°F
Bonnet	Cryogenic	Extended	Standard	Extended	Extended
Packing	PTFE	PTFE	PTFE Chevrons	Graphite	Graphite

Notes

1. On some applications it is advisable to use standard bonnets rather than extended due to vibration considerations.
2. Bellows seal to atmosphere available as an option across all temperature ranges above.

Figure 12 - Leakage Class Specification

Plug Design	Seating Design	Plug Seal Type	Available Leakage Class	Seat
Solid (Unbalanced)	Metal to Metal	None	III, IV, V, VI	Cryogenic to 550 °C (1020°F)
Solid (Unbalanced)	Soft Face	None	VI	Cryogenic to 250 °C (480°F)
Balanced (Plug Seal)	Soft Face	Carb./Graph. PTFE U Seal	V, VI	Cryogenic to 250 °C (480°F)
Balanced (Plug Seal)	Metal to Metal	Graphite Ring	III	250 °C to 550 °C (480°F to 1020°F)
Balanced (Plug Seal)	Metal to Metal	Carb./Graph. PTFE U Seal	IV, V	-100 °C to 250 °C (-148°F to 480°F)
Balanced (Plug Seal)	Metal to Metal	Metallic	IV	250 °C to 550 °C (480°F to 1020°F)

Technical Data

Figure 13 - Standard Material Options for Oil and Gas

Typical Duties	Guide Material	Plug	Stem	Seat
Standard Combination / NACE	316 St.St. + Hard Chrome Plated or 17/4PH St.St.	316L ST.ST. / DIN 1.4404	316L ST.ST. / DIN 1.4404	316L ST.ST. / DIN 1.4404
Sea Water/ Sour Gas	Duplex St.St. & Hard Chrome plating	Duplex St.St./ DIN 1.4462	Duplex St.St.	Duplex St.St.
Sea Water / Sour Gas High H2S Gas	Super Duplex & Hard Chrome plating	Super Duplex / DIN 1.4501	Super Duplex	Super Duplex
Highly Corrosive	Monel K500® Hardened	Monel 400®/ UNS N04400	Monel K5000 / UNS N05500	Monel 400®
Highly Corrosive	Hastelloy C® & hard chrome plating	Hastelloy C®/ UNS N10276	Hastelloy C®	Hastelloy C®
Highly Corrosive	Alloy 625 & Hard Chrome Plate	Alloy 625 / UNS N06625	Alloy 625	Alloy 625
Highly Corrosive / Sea water service	Titanium / Titanium Nitride	Titanium (UNS R50400) / Titanium Nitride	Titanium / Titanium Nitride	Titanium / Titanium Nitride

Figure 14 - Overlay Options for Oil and Gas Applications

Typical Duties	Guide	Plug	Stem	Seat	Temp.	
Cryogenic	Gr.6 Stellite	Gr.6 Stellite	-	-	< -100°C	< -212°F
Low Temp.	Hard chrome Plating	Gr. 6 Stellite or Ultimet	-	-	-100°C to -40°C	-148°F to -40°F
Normal Temp.	Hard chrome Plating	Gr. 6 Stellite or Ultimet	-	-	< 250° C	< 480°F
High Temp.	Gr.6 Stellite	Gr.6 Stellite	-	-	250°C to 400°C	480°F to 750°F
Fast-Acting [> 1" per second]	Gr.6 Stellite	Gr.6 Stellite	Titanium nitride	-	-	-
Liquid pressure drops:- 20 to 30 bar	Hard Chrome plate	Gr. 6 Stellite or Ultimet	-	Gr.6 Stellite or Ultimet	-	-
Liquid pressure drops:- 30 bar to 100 bar	Gr.6 Stellite or Ultimet	Gr. 6 Stellite or Ultimet	-	Gr.6 Stellite or Ultimet	-	-
Liquid pressure drops:- > 150 bar	Labyrinth trim or Tungsten	Gr.6 stellite or Tung. Carbide	-	Gr.6 stellite or Tungsten Carbide	< 150°C	< 300°F
Contaminated Services	Tungsten Carbide Insert	Tungsten Carbide	Tungsten carbide	Tungsten carbide Insert	150°C	< 300°F

Technical Data

Figure 15 - Material Combinations for Chemical and Petrochemical Applications

Typical Duties	Guide Material	Plug	Stem	Seat
Standard	316 St.St. + Hard Chrome Plated or 17/4PH St.St.	316L ST.ST. / DIN 1.4404	316L ST.ST. / DIN 1.4404	316L ST.ST. / DIN 1.4404
Corrosive/ Stress Corrosion Cracking	Duplex St.St. & Hard Chrome plating	Duplex St.St./ DIN 1.4462	Duplex St.St.	Duplex St.St.
Corrosive/ Stress Corrosion Cracking Resistance	Super Duplex & Hard Chrome plating	Super Duplex / DIN 1.4501	Super Duplex	Super Duplex
Highly Corrosive/ Toxic Service	Monel K500® Hardened	Monel 400®/UNS N04400	Monel K5000/UNS N05500	Monel 400®
Highly Corrosive/ Toxic Service	Hastelloy C® & hard chrome plating	Hastelloy C®/ UNS N10276	Hastelloy C®	Hastelloy CO
Highly Corrosive/ High Temperature	Incoloy & Hard Chrome Plate	Incoloy	Incoloy	Incoloy
Highly Corrosive/ Acetic Acid (PTA, PIA)	Titanium / Titanium Nitride	Titanium (UNS R50400) / Titanium Nitride	Titanium / Titanium Nitride	Titanium / Titanium Nitride

Figure 16 - Material Combinations for Power Applications

Typical Duties	Guide Material	Plug	Stem	Seat	Temp. Range
Low-Temp. Boiler	420 St.St. Hardened (1.6142)	420 St.St (Rc 35 to 43) or 17-4 PH St.St. to 43) or (Rc 39 to 41)	431 St.St.	420 St.St. or 316 St.St.	< 250° C (<480°F)
High-Temp. Boiler feedwater	420 St.St. Hardened	420 St.St (Rc 35 to 43) or 17-4 PH St.St.	431 St.St.	420 St.St. or 316 St.St. & Wearth (Colmalloy)	250° C to 350° C (480° F to 660° F)
High-Temp. dry Steam	WC9 (1.7380) & Gr.6 Stellite	WC9 & Gr. 6 Stellite	431 St.St.	WC9 & Gr. 6 Stellite	200° C to 500° C (390°F to 930°F)
High-Temp. Wet Steam	WC9 & Gr. 6 Stellite	WC9 & Gr. 6	431 St.St.	WC9 & Gr. 6 Stellite	200° C to 400° C (390°F to 750°F)
Min. Flow Pump Re-cycle	420 St.St. Hardened or D.L.C. or Inconel	17-4 PH St.St. (Rc 39 to 41)	431 St.St.	420 St.St (Rc 35 to 43)	-40° C to 250° C (-40°F to 480°F)
High-Temp, High-D.P. Flashing Applications	420 St.St. Hardened	420 St.St (Rc 35 to 43) or 17-4 PH St.St.	431 St.St.	420 St.St. or 316 St.St.	< 400° C (< 750 °F)
Turbine Bypass	420 St.St. Hardened for D.L.C. or Inconel	WC9 & Gr. 6 Stellite	Inconel 718®	WC9 & Gr. 6 Stellite	250° C to 400° C (480°F to 750°F)

Figure 17 - Standard Rangeability Values for Cage-Guided Trims

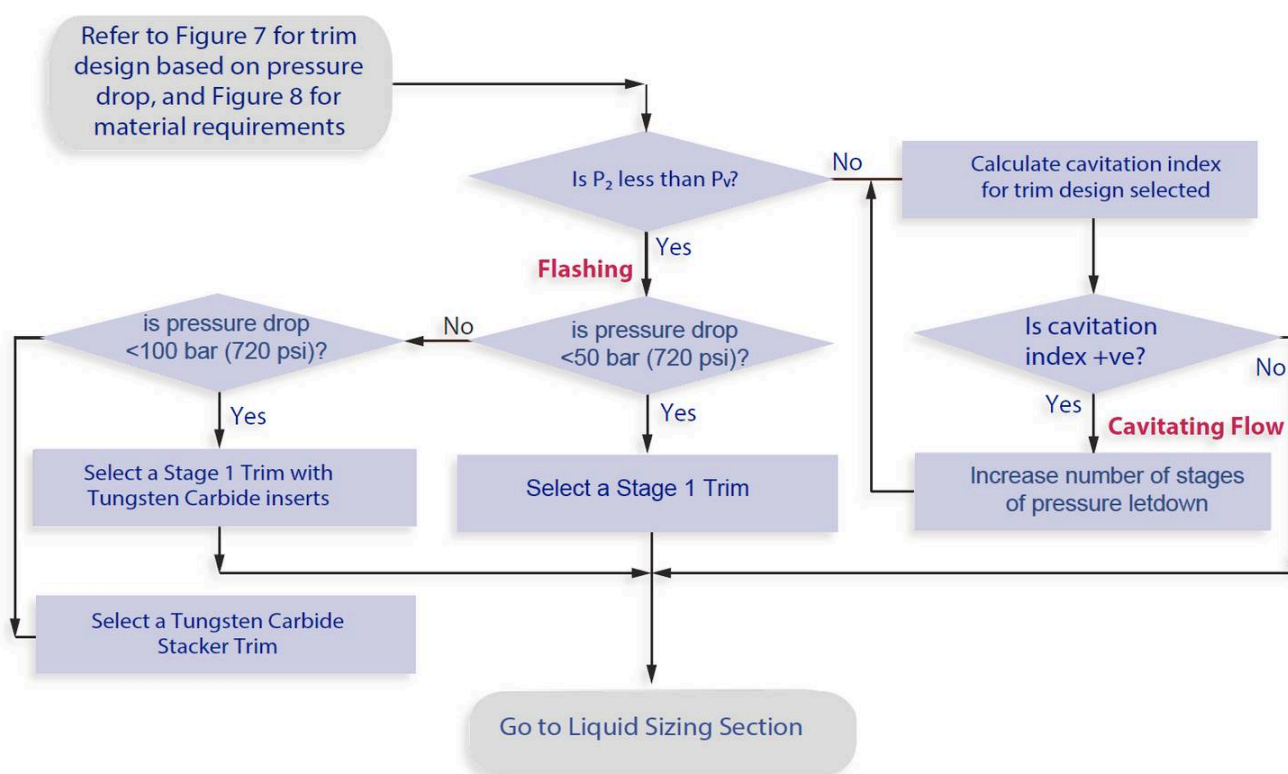
Trim Design	Trims Size Reference	Single-Stage	Multi-Stage
Cage-Guided	< 15mm (< ½")	20:1	15:1
	20mm to 25mm (¾" to 1")	20:1	20:1
	32mm to 80mm (1½" to 3")	30:1	25:1
	100mm to 200mm (4" to 8")	40:1	35:1
	250mm to 350mm (10" to 14")	60:1	55:1
	400mm to 600mm (16" to 24")	70:1	60:1
	> 24"	80:1	70:1
Needle Splines	Cv > 0.75	100:1	90:1
	Cv < 0.75	60:1	50:1

Sizing and Selection Criteria for Liquids

Trim Selection Steps

The chart below presents the steps involved in the selection of a trim for a liquid application. The following sections explain some of the phenomena displayed below and explain the trim designs available to the Application Engineer when selecting the most appropriate control valve design. Reference to the chart below should enable the correct trim to be

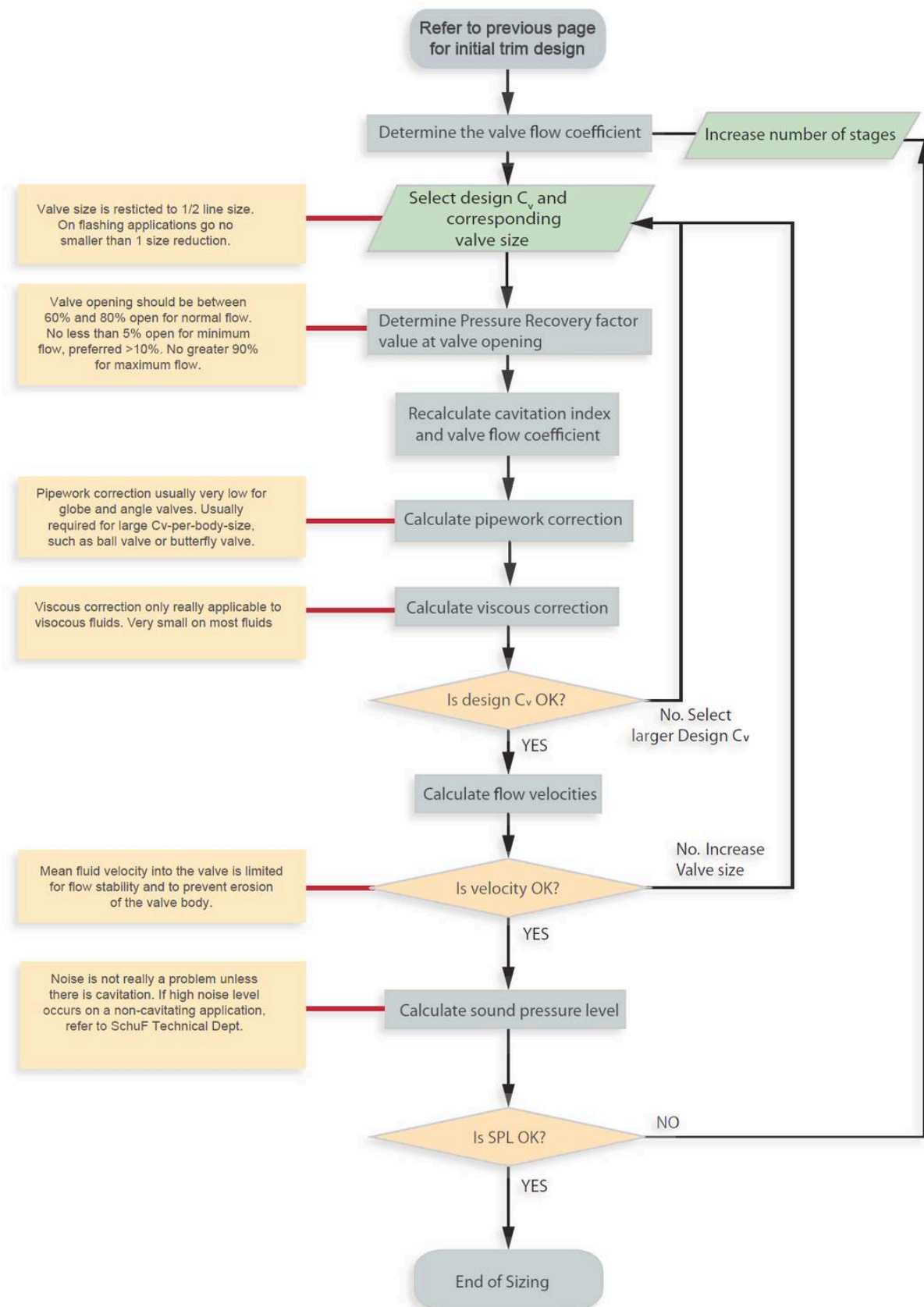
selected for the actual pressure drop and state of the flowing media. However, when working through the flow chart, there may be also a need for an increase in number of stages in order to achieve noise criteria. Usually, if cavitation has been eliminated, noise should not be a major issue.



Right: - Trim components, in a variety of sizes, from 1" to 16"



Sizing and Selection Criteria for Liquids

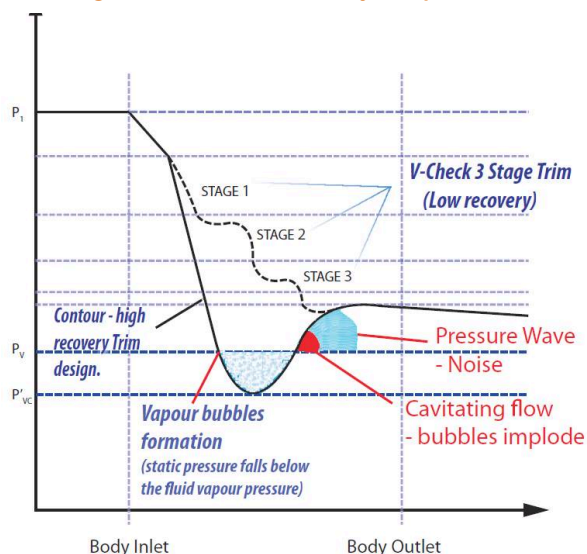


Sizing and Selection for Liquids

Flashing and Cavitation

Flashing occurs when vapour bubbles are generated within the liquid if the local static pressure falls below the fluid vapour pressure. This subsequently results in the phenomena known as **Cavitation**, which occurs if the pressure subsequently increases (recovers) to a level above the fluid vapour pressure- see Figure 17 below.

Below: Figure 17 - Pressure Recovery Comparison



Cavitation occurs when the bubbles previously created during Flashing suddenly implode or collapse. These bubble implosions create micro-jets of fluid and can also create shock-waves and increased noise, which can in turn cause structural damage to the valve components (see Figure 18, above right, which shows cavitation damage to a plug and seat face).

Cavitation can cause damage to the downstream pipework, and must be taken into consideration when valve sizing. Cavitation will also be accompanied by vibration and noise, and the former can, in extreme cases, have a significant impact on flow control, while the latter can be an environmental regulation problem. It is therefore very important to prevent cavitation from occurring. This can be achieved by using multi-stage trims which apportion the



Left - Figure 19a - example of a Multistage Trim arrangement

Right: Figure 18 - structural damage to valve seats (above) and plug (below) as a result of cavitation



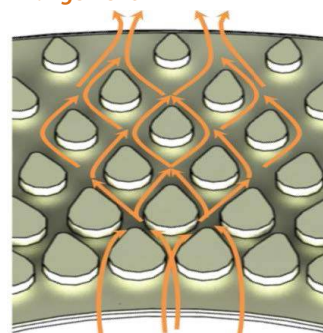
pressure drop across multiple pressure let-down stages within the trim. The chart in figure 17, left, shows the Low-Recovery pressure contour in a three-stage set-up in comparison with a problematic High-Recovery trim contour. Figure 19a (below, left) shows an actual multi-stage trim design. Multi-stage trims make it possible to significantly reduce the potential for cavitation, but the pressure-drop across the earlier stages must also not be high enough to result in cavitation across them.

A concern with multi-stage **Drilled Hole Cage (DHC)** designs is whether or not they do actually apportion the pressure-drop throughout the full length of valve travel. This is definitively achieved by designing the cage to have a large number of galleries throughout the full travel of the valve.

Stacker Trim: Patented Innovation

SchuF have patented unique multi-stage disc-stack trim designs incorporating a series of pressure-reduction strategies which eliminate cavitation, vibration and hydrate / condensate formation. An example of one such complex *tortuous path* flow path is shown in Fig. 19b, right. This disc-stack technology utilises media expansion and contraction, 90° direction-changes without sharp corners, and expanding flow passages to minimise erosion and clogging.

Below: Figure 19b- SchuF- Patented Disc-Stack Flow Path Arrangement



Sizing and Selection for Liquids

Flashing Service/Contaminated service

Flashing occurs when the downstream pressure is lower than the fluid vapour pressure, and there are many applications in the Oil and Gas Industry where this problem is encountered. Experience on flashing applications and multi-phase services has shown that

there can be detrimental performance if multi-stage trim designs are miss-specified. The photographs below give ample evidence of the serious erosion damage that can occur on flashing and/or contaminated services.

Flashing Erosion

The reason for the wear in these cases is attributed to the impingement of high velocity gas / liquid mixtures. This will occur on flashing service or on multi-phase flows, where there is a significant degree of flash or dissolved gas released. This results in a significant increase in the specific volume of the fluid leading to much higher flow velocities and greater erosional forces (see figure 20, right).



Above: Figure 20 - flashing erosion

Contaminated Services

Sand inclusion in flashing fluids can lead to accelerated erosion damage (see Figure 21, right): this photograph gives an illustration of how damaging contaminated services can be. If the incorrect trim design is selected then there is a possibility of breaching the pressure-containing boundary, i.e. the body wall, as in the figure 21 image.

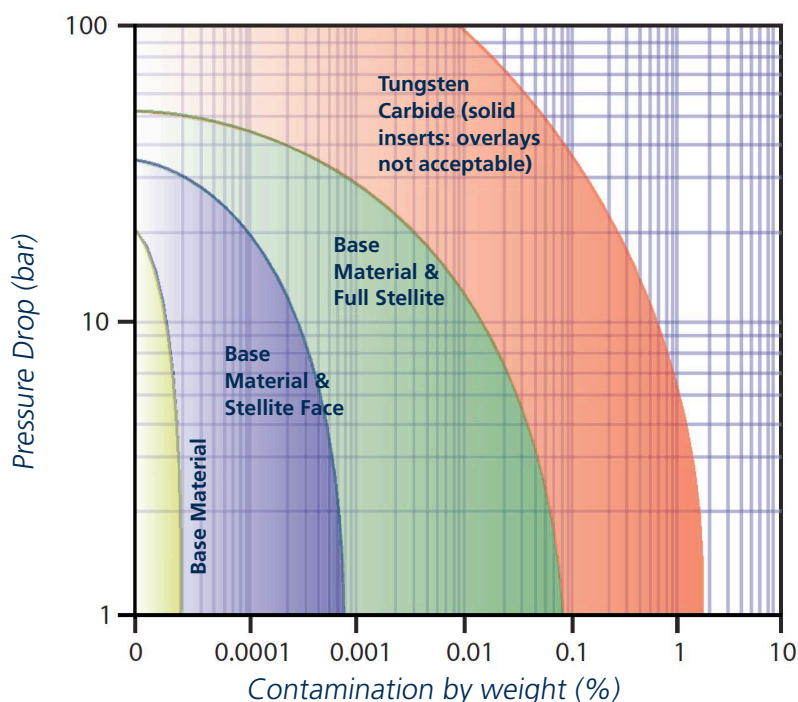


Left: Figure 21 - erosion damage due to contaminated flow

Figure 26 - Material guidelines for Contaminated Services

Carbide coatings should not be used as they are relatively thin, and will erode in time- leading to the base material being uncovered, potentially resulting in severe erosion damage.

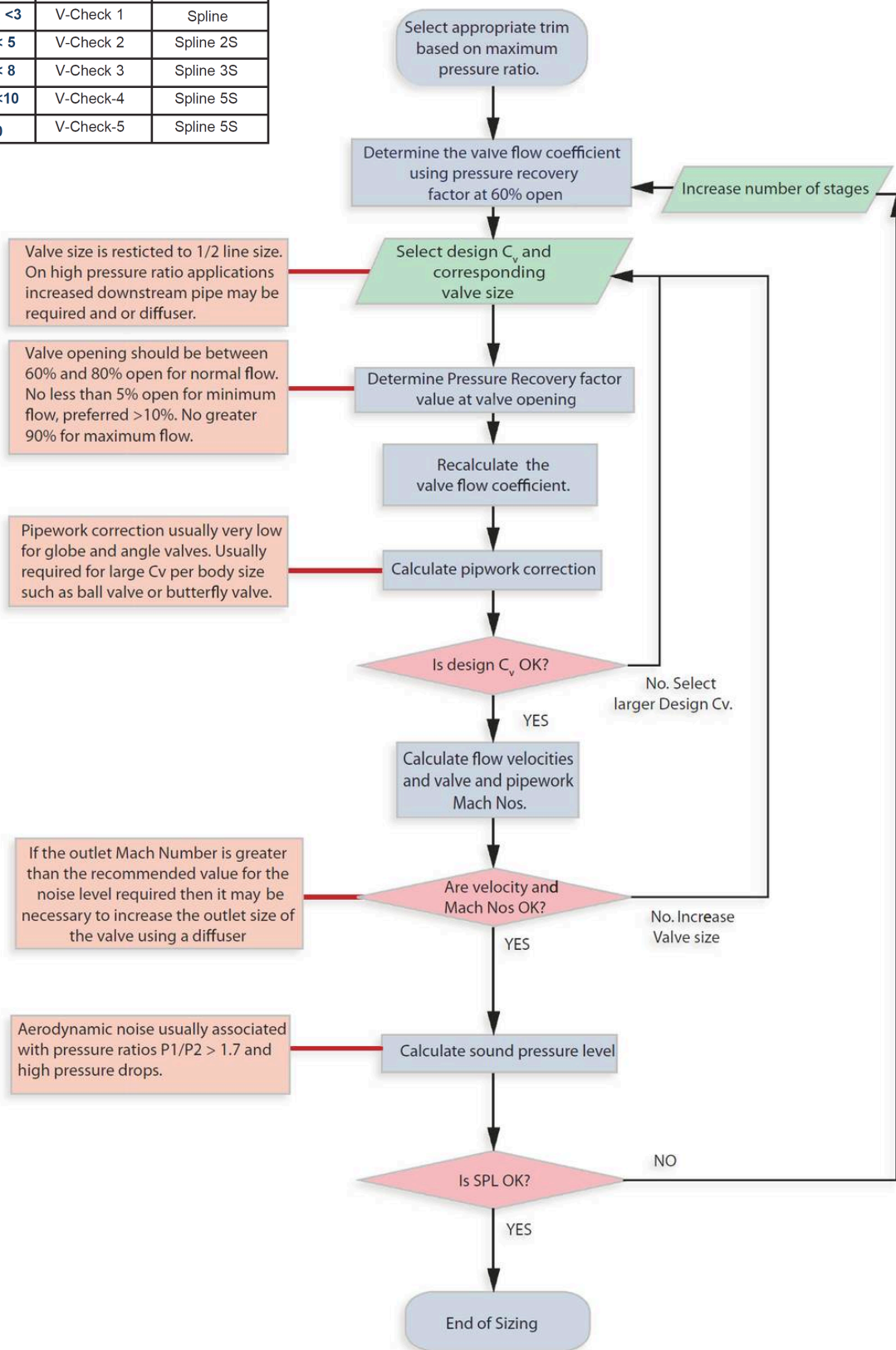
Figure 26 (right) gives guidelines for material selection for contaminated service



Sizing and Selection for Gas / Vapour Flows

Pressure Ratio P_1/P_2	Trim Design	CV <5 Trim Design
$P_1/P_2 < 1.5$	Contour	Spline
$1.5 \leq P_1/P_2 < 3$	V-Check 1	Spline
$3 \leq P_1/P_2 < 5$	V-Check 2	Spline 2S
$5 \leq P_1/P_2 < 8$	V-Check 3	Spline 3S
$8 \leq P_1/P_2 < 10$	V-Check-4	Spline 5S
$P_1/P_2 \geq 10$	V-Check-5	Spline 5S

The table (left) gives an indication only of the trim design required. It must be used in conjunction with the other selection guidelines and calculations.



Sizing and Selection for Gas / Vapour

Aerodynamic Noise Generation

The flow of gas / vapour through a control valve can produce a significant level of noise. The valve itself may function satisfactorily but a by product of the energy dissipated across the valve trim is noise. The noise is generated from high levels of turbulence / high shear, together with compressibility and shock waves. Compressibility effects start to take place as the pressure ratio across the valve i.e. p_1/p_2 increases above 1.5:1. This coincides with compressibility effects starting to dominate.

As the pressure ratio increases to 2:1 shock, wave formation will further increase the noise. In some cases the level of noise generated can lead to vibration problems, both within the valve and the downstream pipework. The levels of vibration can lead to fatigue failure, so it is necessary to both be able to predict when this might occur and find ways of reducing / controlling it.

There are a number of different trim geometries that can significantly reduce the level of noise being generated, which control the flow velocity through the valve trim. The different designs can be classified as DHC (*Drilled Hole Cage*), and labyrinth (*Disc Stack*). Other devices can be incorporated downstream of the valve, i.e. dynamic attenuators / silencers.

Generally the preferred flow direction on a gas / vapour flows is under the plug. This allows the flow area to increase as it passes through the valve trim. Labyrinth type trims will have a large number of stages of pressure letdown each having a relative low area increase per turn.

On a DHC the area increase can be significantly higher. The result is that the flow exiting the final stage of letdown is significantly reduced in both designs resulting in higher noise level attenuation.

Dynamic Attenuators/Diffusers

As well as ensuring the valve is not a major generator of noise, it is also necessary to ensure the pipework downstream of the valve is correctly sized, so that there is no secondary noise generated from the pipework. It is generally accepted that to do this, the flow velocity should be restricted to a maximum of 0.3 of the sonic velocity of the fluid. Flow velocities above this will lead to compressibility affects and high shear resulting in noise being generated at downstream bends, valves etc.

One way of controlling the flow velocity is to use downstream dynamic attenuators, i.e. diffuser plates. These can be designed to control the flow velocity by gradually increasing the flow area thereby reducing the flow velocity at the outlet of the valve. A large number of these units are installed in the Oil and Gas and Power sectors.

SchuF consider all process conditions in the selection of these devices and ensure that the requirements of including "silencers" in a system specification are fulfilled. It would certainly prove easier to operate on the basis that such a system can only be designed to perform optimally at one flow condition, i.e. a large

operating band may cause issues for downstream devices at the lower flows, as they will only produce a minimal pressure drop- but SchuF factor in these issues from the beginning.



Angle Control Valve featuring a diffuser on the outlet connection

Severe Service Trim Designs

4-Stage Cage-Guided Design

The 4-Stage Cage-Guided design, shown in Figure 24, with the sleeves separated, incorporates an inner "guide" and then three outer concentric sleeves. Each sleeve has a multitude of grooves incorporating radial holes. The grooves in each sleeve line up to create a tortuous radial flow path (see illustration Figure 25).

The trim uses the principle of controlled trim exit velocity. The holes within the sleeves are misaligned to produce a tortuous path through the trim. Energy is dissipated within the cage by the combined effect of flow splitting, flow impingement, and turning of the flow as it passes through the sleeves.

There is also a specific increase in flow area between the stages of let down resulting in a reduction in pressure drop as the flow passes from one stage to the next. This significantly reduces the flow velocity, and at the final stage of let down has a much reduced trim exit velocity or velocity head.

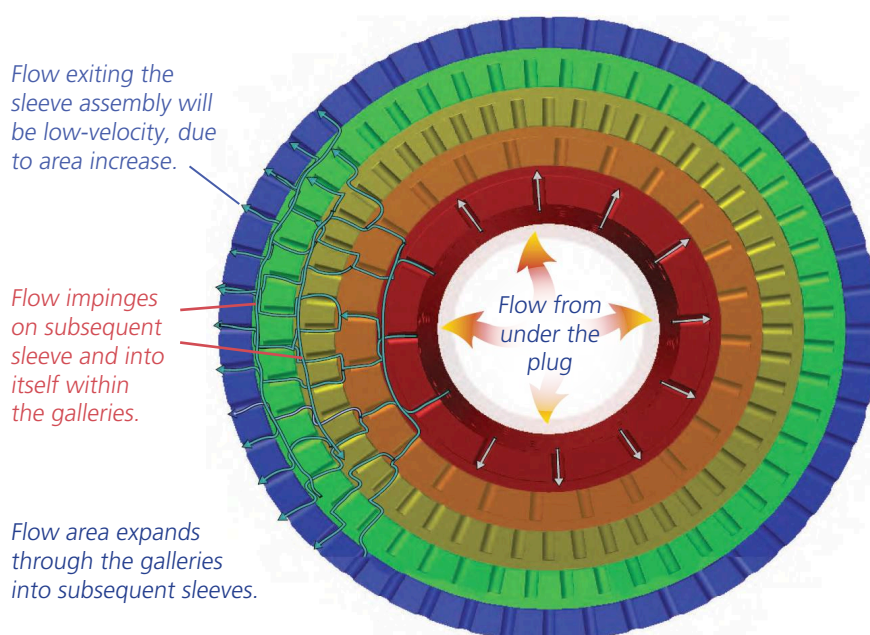
The preferred direction for a multi-stage trim design is under the plug, as this enables the maximum flow

area as the flow passes from inlet to outlet of the trim. On a gas service with a single stage trim the preferred flow direction is also "under" the plug. The main reason for this is that the flow does not recombine as it exits the trim and it has been shown that the acoustic efficiency is significantly lower in this direction.



Above: Figure 24 - A 4-Stage Cage-Guided Design

Below: Figure 25- Multistage Low-Noise Trim, with Radial Flow Path



In cases where further noise reduction is required, smaller holes, i.e. 3mm / 4mm can be utilised in the cage. This can result in further attenuation between 3 to 10 dB(A). This noise reduction is attributed to the smaller scale turbulence structure created by the flow through the smaller holes.

This results in the spectrum of the noise generated being of a higher frequency - which then results in a greater level of noise attenuation from the downstream pipework, and subsequently a lower level of transmitted noise.

Severe Service Trim Designs

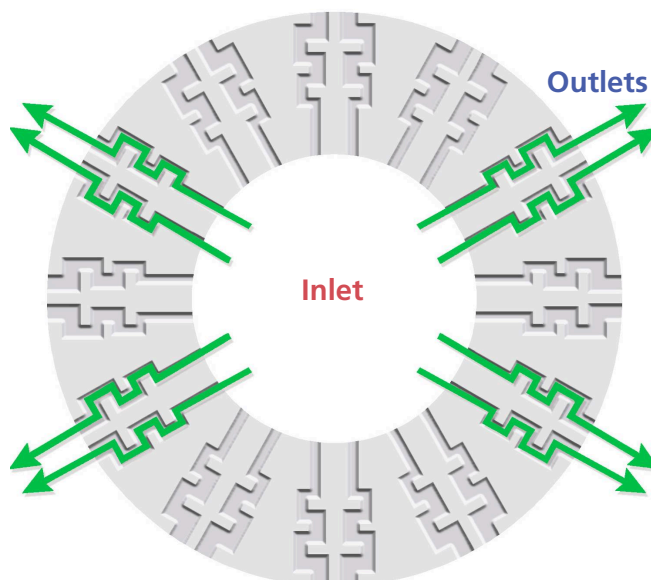
Severe Service Stacker Trim

A further trim option for use on both liquid and gas/vapour flows on what are termed "Severe Service" Applications is the *Stacker Trim*. Whereas there is a practical limit to the number of cylindrical sleeves that can be used on a cage design, it is possible to produce significantly more (30+) on a disc design such as the one shown in Figure 26, right. Although this design has more stages of letdown, the area-increase per stage is relatively small, so the trim exit velocity is not significantly different to a DHC design.

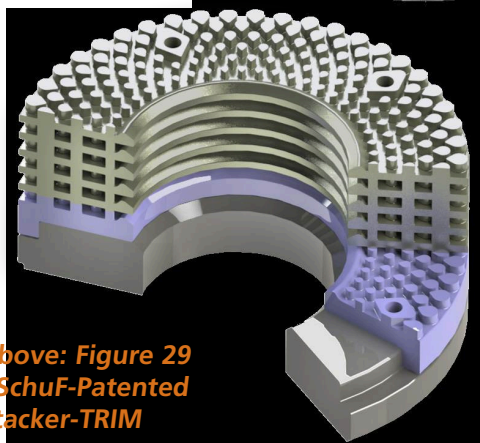
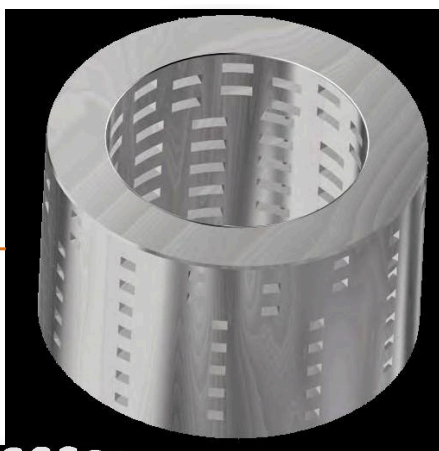


Above: Fig.27 - Stacker Trim in a 600mm (24") valve

Below: Figure 26 - Stacker Trim Flow Path



Right: Figure 28 -
Stacker-TC TRIM



Above: Figure 29
- SchuF-Patented
Stacker-TRIM

There are some limitations to the Stacker design, however; the flow paths are relatively small (< 2 mm), and consequently, on contaminated services, could easily become blocked. For lower pressure-drops on contaminated services, it is possible to use a single-stage tungsten carbide design, but for the extreme pressure-drops being increasingly encountered, it is necessary to introduce further stages of letdown.

The trim presented left, in Figure 28, is made from several tungsten carbide discs. The discs are machined to produce a tortuous path for the fluid, passing radially and circumferentially between each of the adjacent plates. When the tungsten carbide is sintered, these discs become one homogeneous component.

The flow path is a series of 90-degree bends between 2 adjacent plates, in a similar manner as that shown in figure 26 above. The flow area increases after each turn, leading to a reduction in the velocity exiting the disc stack, and thereby reduced erosion.

An alternative disk stack design is shown left. This can provide the benefit of not blocking-up on contaminated services.

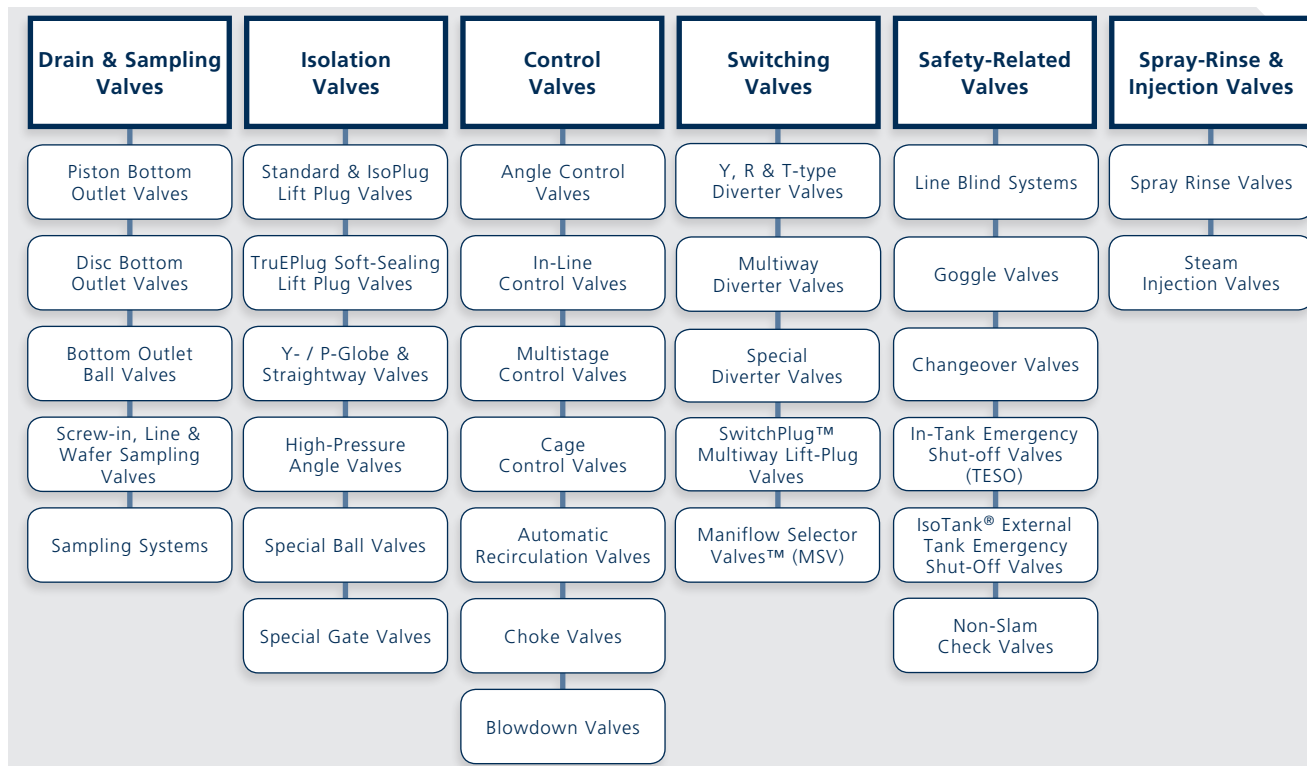
Customer Globe Control Valve Enquiry Sheet

SchuF		CUSTOMER APPLICATION DATA									
		sales@schuf.com					www.schuf.com			Date:	
Customer							Purchase Order No				
Customer Enq. Ref.							Item Number				
Quote Reference							Quantity				
Project Title							Customer Tag				
1	PROCESS DATA	Fluid				Critical Pressure			Shut-off Pressure		
			Units	Min	Norm	Max	Other				
2		Inlet State									
3		Flow Rate Mass									
4		Flow Rate Volume									
5		Inlet Pressure									
6		Outlet Pressure									
7		Inlet Temperature									
8		Molecular Mass									
9		Specific Heat Ratio									
10		Compressibility									
11		Specific Gravity									
12		Vapour Pressure									
13		Viscosity									
14	SPL @ 1m										
15	PIPE	Inlet Pipe Size / Sch				43	ACTUATOR	Actuator Type			
16		Outlet Pipe Size / Sch				44		Actuator Model / Stroke	/		
17		Insulation				45		Size / Spring	/		
18		Design Pressure				46		Fail Action			
19		Design Temp Min/Max.				47		System Air / Set Press.	/		
20	BODY	Valve Type				48	Actuator Orientation				
21		Size / Class				49	Handwheel				
22		Body Type				50	Volume Tank				
23		Body Material				51	Tubing				
24		End Connections				52	Fittings				
25		End Finish				53	Stroking Time				
26		Body F/F Dims				54	Positioner Model				
27	BONNET	Bonnet Type				55	Comms Protocol				
28		Bonnet Material				56	Control Signal				
29		Bonnet Bolting				57	Switch Type				
30		Gasket Material				58	Filter Regulator				
31		Gland Packing				59	Boosters				
32		Bellows Material				60	Solenoid				
33	TRIM	Trim Size / Type				61	Lockup Valves				
34		Cv / Characteristic				62	Relief Valve				
35		Balancing				63	Other Accessories				
36		Flow Direction				64	Other Accessories				
37		Plug/Seat/Cage Matl.				65	TEST	Design and Testing: Seat Leakage, Hydrostatic, Functional Test			
38	Seating Type				66						
39	Seat Leakage				67						
40	CERT	PED				68	MISC	Paint Finish	Standard	☐	
41		NACE				69		Mat. Cert Body / Bonnet	/		
42		ATEX				70		Drawing Outl / GA / Hkp	No / No / No	☐	
71	NOTES								Rev.	Date	Desc.
72											
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Product Portfolio Overview

The SchuF Group has delivered over one million valves during its 100-year history, to a wide variety of industries in over 50 countries worldwide. With headquarters near Frankfurt in Germany, the company has additional design and manufacturing centres in Italy, India, Ireland, UK and the

USA. SchuF has sales and agent offices servicing almost every country in the world. SchuF manufacture valve products that control, isolate, divert, and sample liquids, gases, powders, and slurries. Our product range of engineered, customised valves includes:



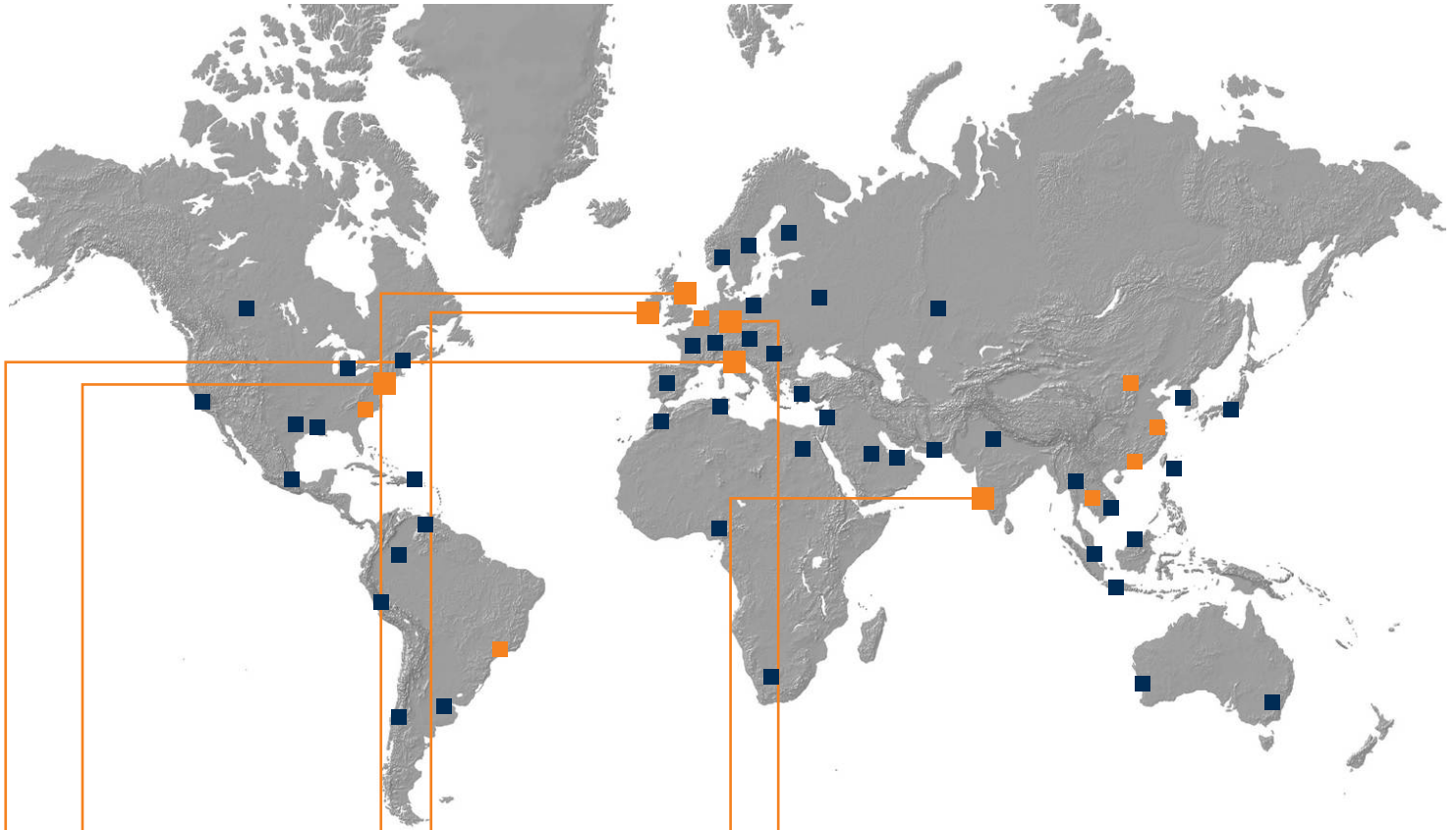
Key Client List:

- | | | |
|------------------|---------------------|------------------|
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| ■ AkzoNobel | ■ INEOS | ■ Roche |
| ■ AstraZeneca | ■ Invista | ■ SABIC |
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| ■ DOW Chemical | ■ Lukoil | ■ Shell |
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| ■ Eastman | ■ Merck | ■ Sinopec |
| ■ Evonik | ■ Novartis | ■ Sulzer |
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